frozen | trained | proposal | SKU prediction | gradient

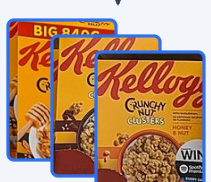
(a) Class-agnostic proposals



shelf image, 1 920 × 1 440 px

prompt: "product . box . pouch ."

Grounding DINO / YOLO-World
open-vocabulary detector



N crops x_i with scores s^{det}
(one per product facing)

(b) Frozen embeddings

DINOv2
ViT-B/14

f_i

768-d

reference gallery: 1 to 5 images per SKU



Clusters honey | Clusters choc. | Orig. regular | Orig. 840 g

one brand, four SKUs: hard negatives

DINOv2
shared weights

e_k

"Kellogg's Crunchy Nut Clusters, chocolate"

CLIP text + image

t_k

(c) SKU adapter and re-ranker (ours)

SKU adapter g_θ

residual MLP, 768 → 256
applied to f_i and e_k

$cos(z_i , \hat{z}_k)$

crops

.82	.71	.30	.22
.66	.79	.25	.20
.28	.24	.74	.69
.21	.27	.62	.81

gallery SKUs

same-brand blocks

Fusion re-ranker

$s_{ik} = w_v \cos(z_i , \hat{z}_k) + w_t \cos(c_i , t_k) + w_d s^{det}$

3 weights + temperature τ; top-k SKUs, reject if max s < δ

Brand-aware contrastive loss

$\mathcal{L} = -\log [\exp(s_{i,y}/\tau) / \Sigma_k \omega_{ik} \exp(s_{ik}/\tau)]$

$\omega_{ik} = \gamma > 1$ if SKU k shares the brand of the true SKU y, else 1

trained on base SKUs; novel SKUs need only a gallery image

(d) SKU-level output



Top-3 for the chocolate crop



text prompt only
Crunchy Nut, orig.
Clusters, honey
Clusters, choc.

ours (re-ranked)
Clusters, choc.
Clusters, honey
Original, regular

scores are illustrative, not results

Metrics: SKU top-1 / top-5,
SKU-level mAP_{50}, novel-SKU accuracy